



WESTMINSTER ASTRONOMICAL SOCIETY, INC. (WASI)



Membership News



Volume 42 – Number 2

Spring 2026

Message from our Society's President, Brooke Parkhurst

Welcome to summer! I hope we'll see you at one of our many summer events. Besides just being fun, your presence supports belief in science—which is more and more important. And anyway, you will definitely have a fantastic time, meet really friendly people, and have fun geeking out about astronomy with like-minded peers.

We look forward to seeing you!

Notes and Thoughts

WASI is often the number one astronomy club in the USA for doing outreach events. In July we're especially busy. The week of July 6 sees us at the Space Themed Vacation Bible School (at St. James in Mt Airy), the Winfield Volunteer Fire Department Carnival, and Shore Leave in Lancaster, PA. Want to meet the public and convey your passion for astronomy? These are great opportunities for that, and we could use some more volunteers. Please let Skip (skipastro@gmail.com) know if you can help. And, as always, you can see all of our activities on the calendar (<https://nightsky.jpl.nasa.gov/events/wasi/>).

July 4 is a special day for various reasons, not the least is a fun chance to spot Uranus and Mars. The two planets will be a mere 7 arcminutes apart in the morning twilight—that's twice as close as the famous Mizar and Alcor pair in the Big Dipper. You might be able to spy Uranus with binos; a small telescope will surely (sans clouds!) show this event.

Pluto can be a tough target, but it will reach opposition July 27. Late at night it will be above the horizon. For a few days it will peak at altitude 27 degrees around 1 AM.

Our annual picnic will be Saturday August 1 at the pavilion at Bear Branch. It's a potluck, so let us know what you'll be bringing by entering that data here: <https://westminsterastro.groups.io/g/main/table?id=37370&lv=&p=...20,0,0,0>

This newsletter goes out quarterly. Do you have thoughts, insight or useful information for the members? Please consider submitting an article. Our survey last year showed that members really want more learning resources, so your expertise could be valuable to many. Send your ideas to secretary@westminsterastro.org.

What Does Hydrogen Alpha Mean?

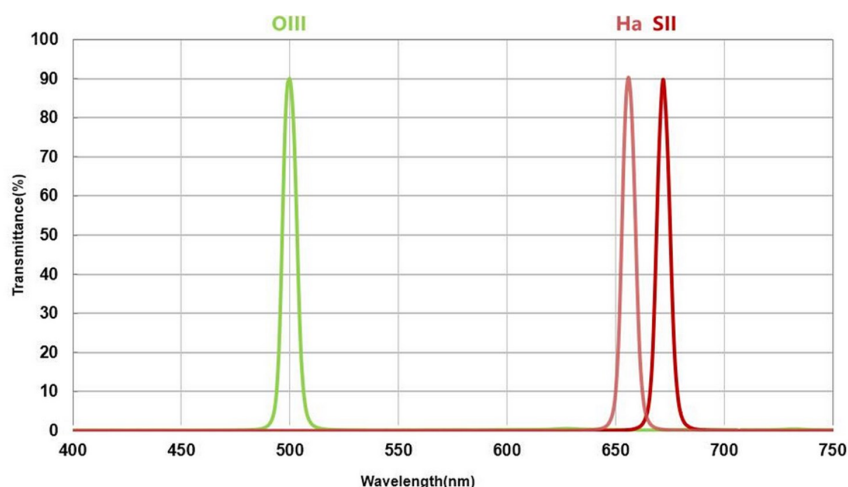
All astrophotographers use monochrome sensors. If you have a color (AKA “One Shot Color” or OSC) camera, the sensor itself is mono, but a filter matrix covers that to separate out the red, green and blue components of starlight.

Some of us use narrowband filters, often called “NB imaging”. Though the pros will employ a huge variety of these, we amateurs mostly use hydrogen alpha (H α), oxygen III (OIII) and sulphur II (SII) filters. Each of these pass a very narrow slice of the spectrum: H α at 656 nanometers (red), SII at 672 nm (also red) and OIII at 501 nm (blue-green).

Why do we use a narrow slice of the spectrum?

By selecting these specific wavelengths we can grab certain aspects of objects while rejecting things like light pollution. In postprocessing it’s common to use this data to highlight aspects of what we’re observing.

The universe is made up of hydrogen (H) and helium. Everything else, like the silicon that makes up sand and dirt, and the carbon so needed for life, is relatively rare.



Hydrogen is the simplest element, consisting of just a single proton and one electron. But that electron can be in one of many energy states. Remember learning that an atom looks like the solar system, with the nucleus at the sun’s position and electrons sedately orbiting like planets? That model is completely wrong; electrons move in orbitals in often complex ways (and, indeed, to speak of an electron’s *position* is somewhat naïve). But the planetary model is a useful simplification at times.

Electrons have energy; the lower the energy the closer (using the simplified model) it is to the proton. They’re a bit on the lazy side and prefer to exist in the lowest energy states.

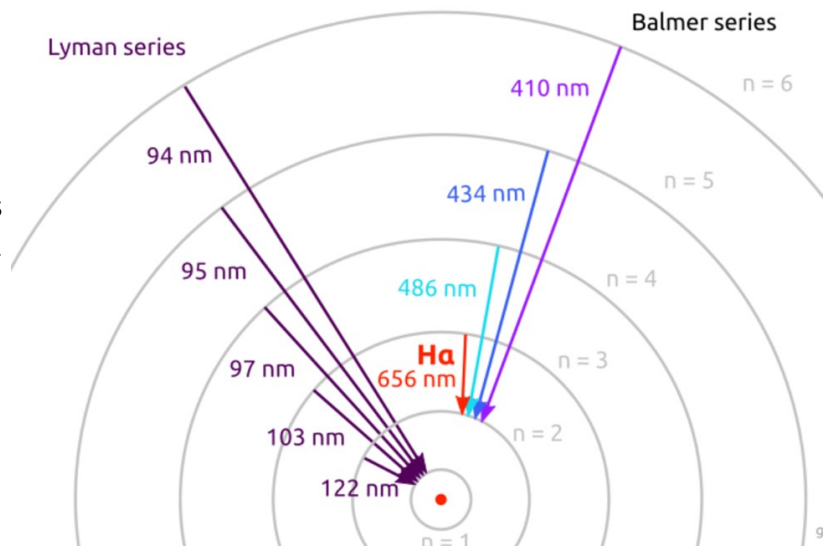
The universe is a violent place. A nebula likely has a hot star at the center emitting all sorts of energetic photons and other particles. If a hydrogen electron is hit by a photon of the right energy, it will absorb that and move to a corresponding higher energy level.

Preferring those lower levels, the energized electron will spit out a photon, losing energy, to descend “closer” to the proton. Energy is always conserved, so that photon’s energy exactly matches that lost by the electron.

What Does Hydrogen Alpha Mean? - Continued

This drawing illustrates these transitions. The innermost “orbit” is denoted by “ $n=1$ ”. The letter “ n ” is called the principle quantum number, and is just the energy state. As n gets bigger the electron has a higher energy.

The purple lines on the left show electrons going from various levels to the innermost, $n=1$. Each change releases some energy. For instance, going from level 2 to 1 releases a photon whose energy results in a wavelength of 122 nm. From 3 to 1 emits one at 103 nm, etc. Any transition to the $n=1$ is a part of the Lyman series. And none of these are of interest to amateur astronomers as they are all in the ultraviolet—we can’t see them, nor can our cameras detect them, and, to make matters worse, the atmosphere blocks them¹.



The Balmer series is more fun. This is the set of transitions to the $n=2$ level. An electron going from 3 to 2 emits a photon at 656 nm. It’s called hydrogen alpha. The next is $n=4$ to $n=2$, and that’s hydrogen beta, working through the Greek alphabet.

The Lyman and Balmer series are the tip of the iceberg; many, many more transitions are possible. Most emit photons we can’t see.

Hydrogen alpha is special because we *can* image those photons, and H α is very common. When a central star pounds an H atom with photons sometimes electrons will absorb those and go to higher levels. It could be $n=3$, which will be followed by an H α photon we can image. Or, if the incoming photon has a higher energy, that electron could be boosted to the $n=4, 5, 6$ —you name it. It will cascade to lower levels, likely including H α , showering the universe with photons at energies shown in the diagram. And giving us photos like this, where the red is glowing hydrogen gas undergoing H α transitions.



What about those OIII and SII filters? The mechanism is more complex... and is actually prohibited by the selection rules of quantum mechanics! It’s impossible to create these transitions in a laboratory because we can’t generate a vacuum approaching that found in space. The poor lab vacuums result in plenty of particles zooming around that can knock electrons out of these “forbidden” states before they emit a useful photon.

¹. Actually, we can detect some of these photons. Quasars emit a lot of very high energy photons, for example at the Lyman alpha wavelength of 122 nm. Because the universe is expanding, red shift will stretch these out to longer wavelengths, often in the visible part of the spectrum.

What Does Hydrogen Alpha Mean? - Continued

Yet these do happen as quantum mechanics never says never; it's all about probabilities. The OIII and SII emissions require more explanation than can be offered here, but you can get the details here: https://www.ganssle.com/misc/ha_oiii_and_sii_transitions_explained.pdf.

And what about those OIII and SII symbols? OIII simply means an oxygen atom that is missing two electrons. SII is a sulphur minus one electron. Shouldn't these symbols then be OII (missing two) and SI (missing one)? This nomenclature is a bit tortured as the Romans never got around to inventing a Roman numeral for zero. (They used the Latin word *nulla* to indicate nothingness). So OI is an oxygen atom missing zero electrons.

A neutral H atom is often called HI and one ionized (missing its electron) is HII. But the terms H α , H β , and the like let us know *which* transition we're talking about when working with HI atoms.

For those curious about the physics: why does H α emit at 656 nm? At $n=1$ the energy of a hydrogen electron is -13.6 electron volts (eV). An eV is an incredibly tiny amount of energy; the energy to lift an apple one meter is roughly 10^{19} eV. The formula for the energy lost or gained in a hydrogen electron transition is:

$$E = -13.6 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Where n_1 is the first energy level and n_2 the second. For the H α transition, $n_1=2$ and $n_2=3$, since the electron is falling from the $n=3$ to $n=2$ level. Doing the math, $E = -1.889$ eV. We know that a photon's wavelength (λ) is:

$$\lambda = \frac{hc}{E}$$

Where c is the speed of light (299,792,458 meters/second) and h is Planck's constant (4.136×10^{-15} eV-seconds). Using the energy just computed, we get:

$$\lambda = \frac{4.136 \times 10^{-15} \text{ eV} \cdot \text{s} \times 2.99792458 \times 10^8 \text{ m/s}}{1.889 \text{ eV}}$$

$\lambda = 656 \times 10^{-9}$ meters, which is 656 nm. (Notice how all of the units, other than meters, cancel out).

You can do this calculation to get the wavelength of any electron transition of hydrogen electrons.

(Since the lowest energy state of a H electron is -13.6 eV, if an incoming photon of 13.6 eV hits the electron it will escape and the atom will be just a lonely proton. Any atom missing an electron is called "ionized", and for hydrogen that's called HII. There are many HII regions in space, often caused by a very hot O or B star's photons stripping electrons away from HI clouds.

Globular Clusters

Globular clusters are tight groupings of stars that are gravitationally bound in a spherical organization. They have thousands to millions of stars in a compact form. Omega Centauri, our Milky Way's biggest, has 10 million stars packed inside of 160 light years. Alas, that gem isn't visible from Maryland, but many others are.

The best is M13, Great Globular Cluster in Hercules. This puppy is magnitude 5.8, possibly viewable by naked eye in dark conditions. With a telescope it's a delight not to be missed. It packs about a half million stars. It is one of 29 globulars in Messier's catalog.

The Milky Way has over 150 known globulars, compared to an estimated 500 to 1000 in M31. It's possible our galaxy is equally verdant, but most are invisible to us due to our position. The giant elliptical galaxy M87 has something like 13,000 of these objects.

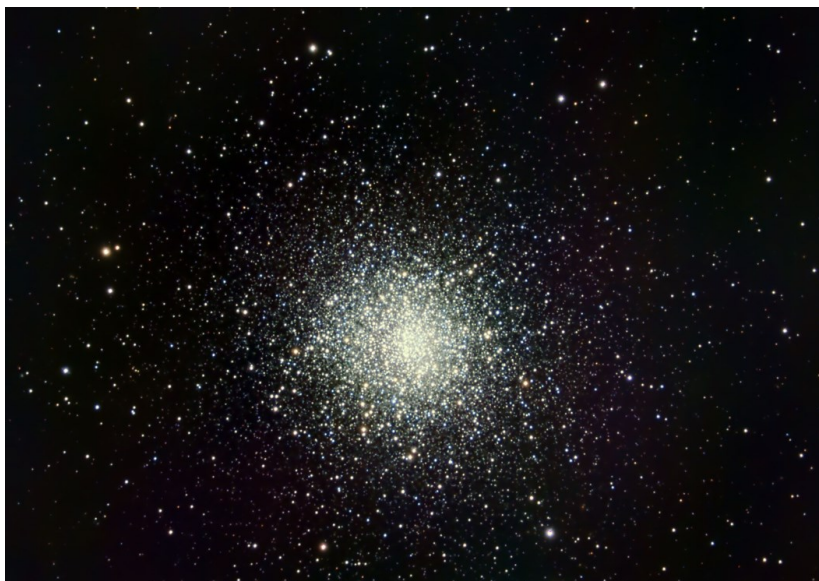
It's not clear how these objects form.

They're mostly very old, often the oldest objects in a galaxy, some nearly 13 billion years not-so-young. As such, many of the stars within are population II, which means they have few heavy elements (any element heavier than helium is called a "metal", and globulars are metal-deficient). Our sun is a population I star, formed after the universe was seeded with metals formed in older stars, disbursed by supernovas. It's thought that some globulars are remnants of small galaxies that were captured by larger ones.

It seems logical to propose that these stars all formed about the same time from a single gas cloud, and, indeed, that was once believed. But more recent observations show at least several generations of stars in globulars.

This image of M13 makes one wonder about the star density at its core. Are they all banging into each other? It seems not. At the core the stars are a few thousand astronomical units apart, around 100 times closer than Proxima Centauri is to the sun. If we were on a planet there the nearest neighbors would be hundreds of times brighter than Sirius.

There are some interactions. Blue straggler stars are hotter and more luminous than expected, and some think that in these clusters they're the result of stellar mergers. And very rapidly-spinning pulsars (neutron stars emitting radio waves) are unusually common, probably because when the clusters formed long ago, there were many very massive stars that went supernova. Neutron stars result from some supernova events.



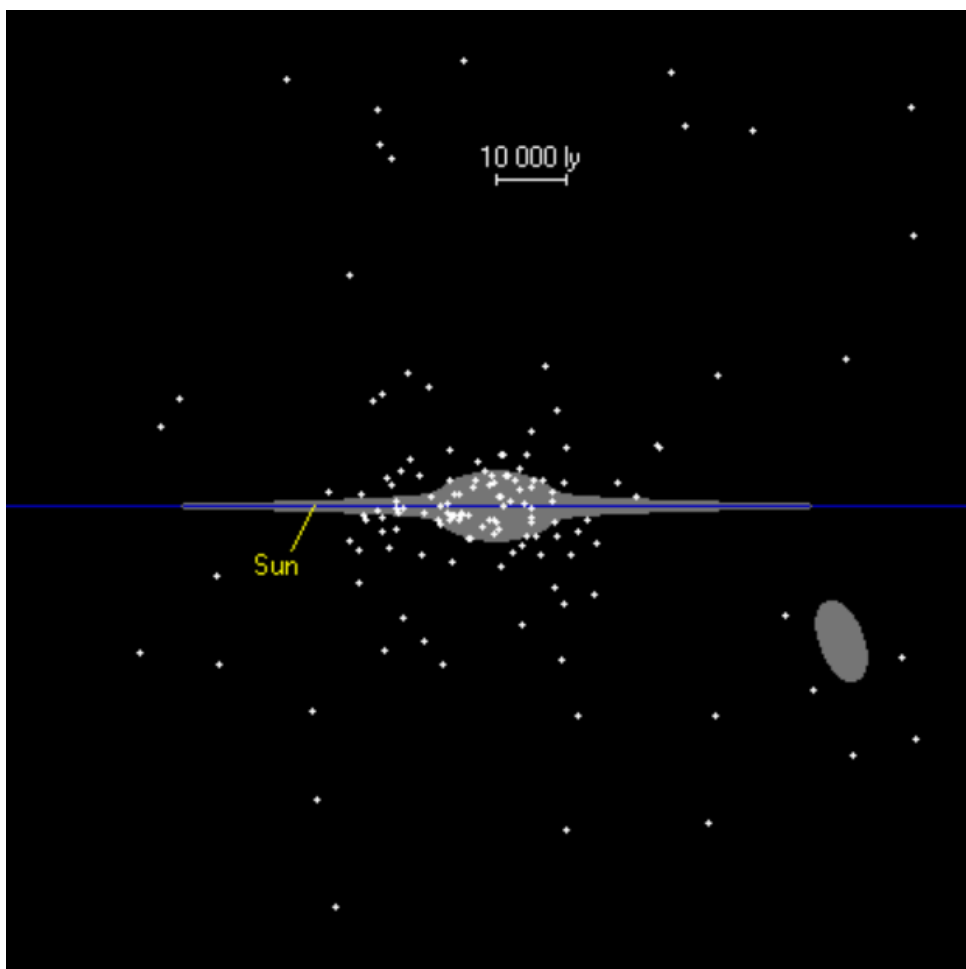
Globular Clusters—Continued

Other supernovas can produce black holes, and it's thought the clusters have plenty of stellar-mass BHs. What about really massive ones? The evidence is scarce, but many think that at least intermediate-sized BHs are found in globulars, and perhaps supermassive ones exist as well.

Strangely, these lovely objects are mostly found in galactic halos, orbiting above and below the planes of galaxies. Some even have retrograde orbits, meaning they swing around in a direction opposite to the galaxy's rotation.

So, next time you're under the stars, check out some of these objects. M13 is to die for. M15 is one of the densest known. M92, like M13 in Hercules is one of the oldest globulars with a highly concentrated center.

Matt Orsie recently shared a link with interesting info about globulars: <https://bigthink.com/starts-with-a-bang/globular-cluster-terzan-5-imposter/>.



The dots are Milky Way globular clusters. Notice how they “cluster” in the galaxy’s halo. Image courtesy of <http://www.atlasoftheuniverse.com>

Astrophotos From Our Members

Laurie Ansorge sent this stunner of M101, the Pinwheel Galaxy. 72 minutes from her Celestron Origin, no post-processing.



And here's her 29 minutes on
M51

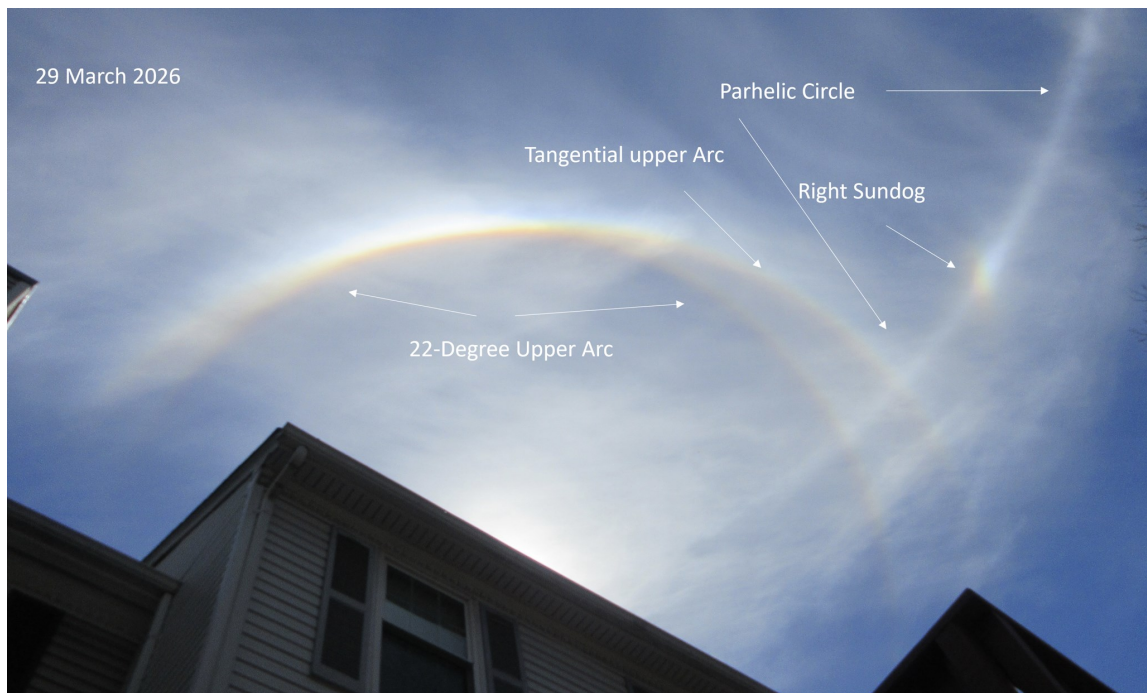


Astrophotos From Our Members

From Jack Ganssle: A friend is on sabbatical in Peru. I helped him a little with his new book “First Principles of Astrophotography” (Boyd Edwards) and in return he offered to image something southern for me. He sent 100 GB of Eta Carina data from atop a 13,000 foot mountain. Here’s my take using the HOO palette with RGB. 10 hours with a Williams Optics UltraCat 108 and an ASI 6200MM.



Jeff Kretsch is a very dedicated solar observer. Check out these artifacts:



Astrophotos From Our Members

From Matt Orsie: The Necklace Nebula is a very small (approx 21"x15" arc sec) planetary nebula in the small northern constellation Sagitta the Arrow. Cataloged as either IPHASX J194359.5+170901 or PN G054.2-03.4, it's a lot easier to refer to it by its common name, the Necklace Nebula. **I'm pretty sure this is the only image anyone has of it taken with a Seestar, S50 in this case.** 3x drizzle and 250% crop. I observed this visually in 2020 with my 22" telescope. Researchers in the Canary Islands, Spain, discovered it in 2005 in data acquired by the Isaac Newton Telescope Photometric H-alpha Survey. The Necklace Nebula lies about 15,000 light-years away and has a diameter of about 2 light-years. This planetary nebula formed thanks to a pair of Sun-like stars in a close binary system at the center. One of the stars evolved a bit faster than the other, becoming a giant. About 10,000 years ago, the giant star expanded to a point where it completely engulfed its companion, called a "common envelope." But it didn't consume it. Because the outer layers of the giant were composed of such thin gas, the Sun-like star continued to revolve around the center of its captor. This dramatically increased the giant's rotational speed. Eventually, the larger star spun so fast that a large percentage of its outer atmosphere expanded into space, producing the planetary nebula. In this case, most of the gas flew outward along the giant star's equator, producing the ring we now see. The bright spots along the ring are regions of denser gas. The Sun-like star continues to orbit the core of the giant about every 27 hours. And the two stars are close, separated by only about 2 million miles (3.2 million km).



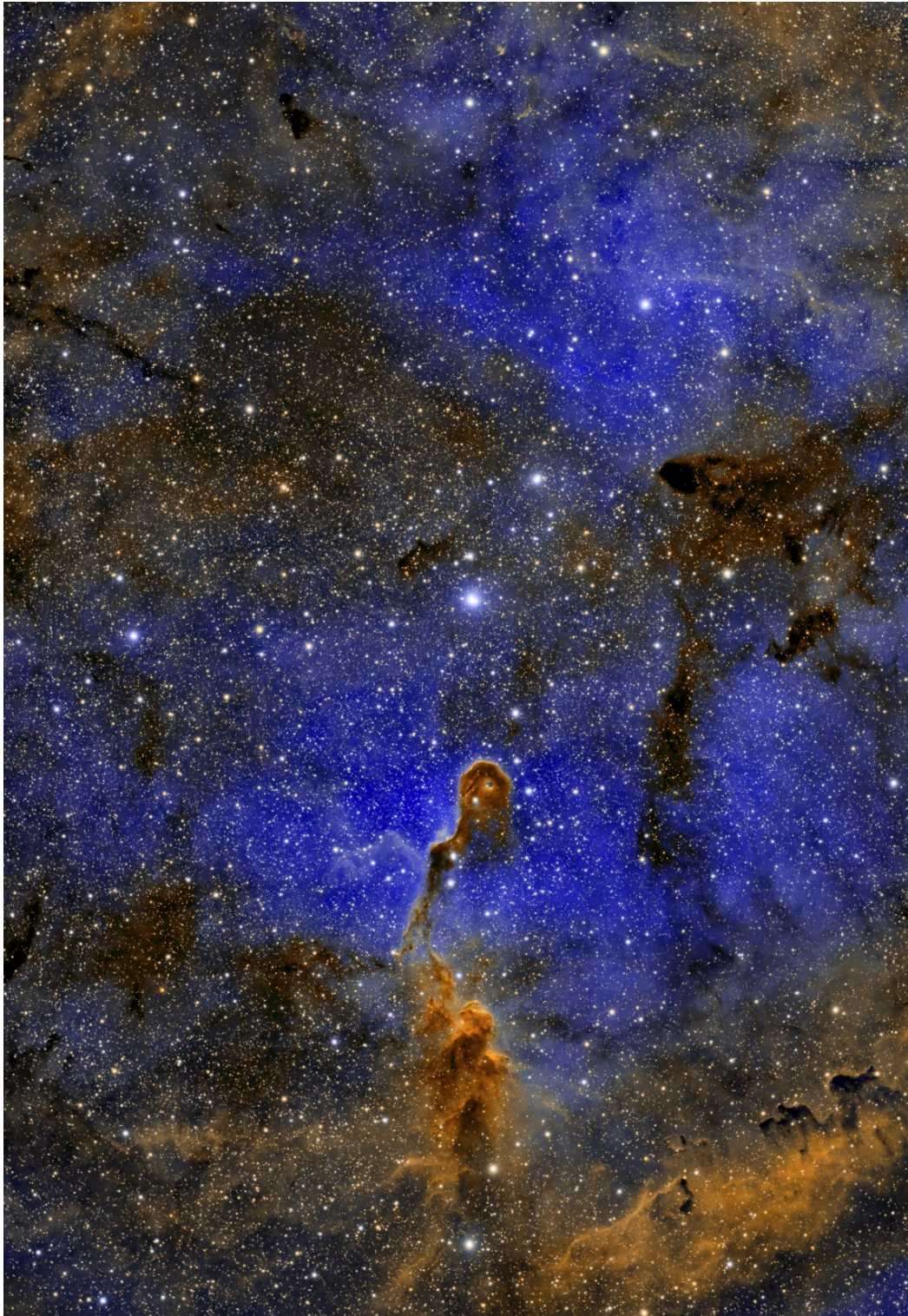
Astrophotos From Our Members

Matt continues: The Abell 39 planetary nebula was cataloged by George Abell from the Palomar Sky Plates taken in the 1940's. His catalog has some of the fainter larger planetaries noted in the Milky Way. Number 39 is one of the most symmetrical that I have observed and photographed. It's 3 mins in diameter at around 13th mag. Here is 9 hrs taken over a couple of recent nights with the Seestar S50 from my Bortle 4.5'ish skies. Post processed in Siril and Photoshop. Of note are a couple of background galaxies seen through the nebula itself. The one at 2:00 about 2/3rds of the way out from the center is approx 17th mag and the dim dot on the limb at 6:00 is approx 18th mag.



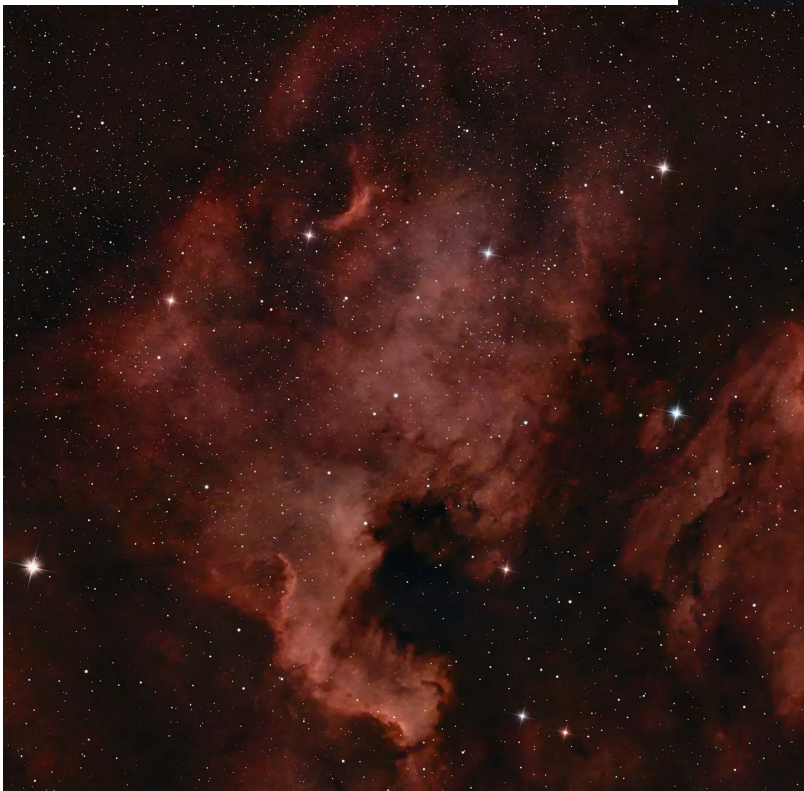
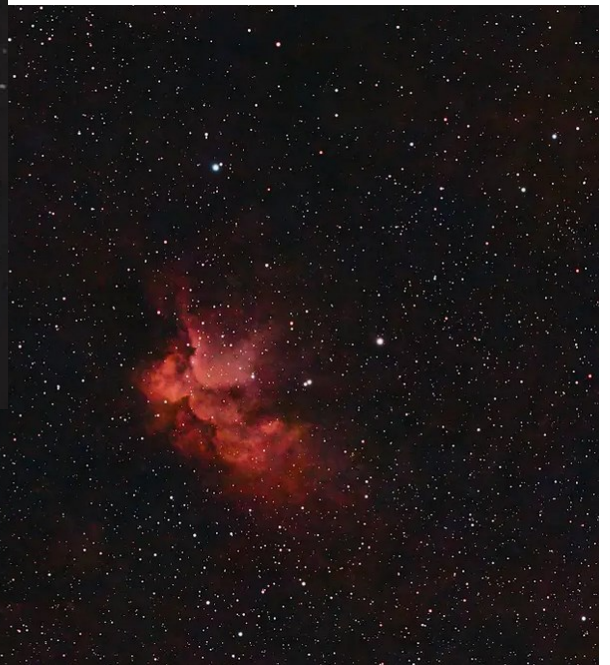
Astrophotos From Our Members

Mikey Mangieri took this of IC 1396, featuring the Elephant's Trunk Nebula. Telescope: William Optics Gran Turismo 102 / GT102, Camera: ZWO ASI2600MM Pro, Mount: Astro-Physics 1100GTO, Filters Astrodon H-alpha 5nm, 31 mm Astrodon OIII 3nm 31 mm, Astrodon SII 5nm 31 mm, 26 Hours integration time



Astrophotos From Our Members

Suhas Hg captured these with his Dwarf 3:



WASI FAQs

Library - Did you know we have over 700 books about astronomy in our WASI library? There are available to WASI members. Here's the complete card catalog: <https://westminsterastro.groups.io/g/main/files>.

Loaner telescopes - We also have a telescope lending library. If you'd like to borrow a scope, talk to Curt Roelle.

Astronomical League - All WASI members are also members of the Astronomical League. Check out their 80+ observing programs, many of which come with awards: <https://www.astroleague.org/>

Newsletter - Send pictures, articles, and ideas for the newsletter to secretary@westminsterastro.org.

Facebook - We're active and sharing images on our Facebook page, found here:



Join/Renew membership link: <https://www.westminsterastro.org/join-wasi/>

If you've already entered your contact information (renewing), skip the "database" link: <https://paypal.me/WAstroSInc>

Dues are payable via PayPal on the link above, by check or cash (and through your bank's on-line bill payment). Membership Dues are \$25/year for individuals or family, and youth under 18 is \$5/year. Or, save money and sign up for 2 years (\$40), or 4 years (\$80).

- On time payment means eligibility for the annual incentive.
- Access to the club's observatory (after training).
- Keep access to the members-only groups.io pages/information
- Receive members-only access/notifications on Night Sky Network
- Keep/get discount rates for popular astronomy magazines
- Borrow from the WASI scope/literature library

Files and club member correspondence & wiki links are found here: <https://westminsterastro.groups.io/g/main>. Remember to set your communication preferences.

Outreach/event calendar is found on: <https://nightsky.jpl.nasa.gov/index.cfm>. Set your communication preferences here as well.

Changed address, email or phone? Please update your information and send a message to the webmaster and/or treasurer@westminsterastro.org.

We meet monthly on the 2nd Wednesday of the month:
Back to Basics from 7:00 PM – 7:30PM; General Meeting 7:30PM – 9:30PM
Bear Branch Nature Center Carroll County; 300 John Owings Rd.; Westminster, MD 21158
Website: <https://www.westminsterastro.org/> (Zoom info for hybrid meetings)